Internet of Things - Project 3

Cloud-Connected Security Node (IoT Telemetry)

Objective

To implement a cloud-connected security node using an ESP32 and HC-SR04 ultrasonic sensor that publishes real-time telemetry to a cloud dashboard.

Implementation

- Interface HC-SR04 with ESP32.
- Connect to Wi-Fi.
- Measure distance.
- Upload data to cloud.
- Display intrusion status.

Source Code

```
#include <WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"

#define WLAN_SSID "Wokwi-GUEST"
#define WLAN_PASS ""

#define AIO_SERVER "io.adafruit.com"
#define AIO_SERVERPORT 1883

#define AIO_USERNAME "Sagana127_"
#define AIO_KEY "aio_ADQL180mLHL37ewwzLenVAi6TfXV"

#define TRIG_PIN 5
#define ECHO_PIN 18

WiFiClient client;

Adafruit_MQTT_Client mqtt(
  &client,
  AIO_SERVER,
  AIO_SERVERPORT,
  AIO_USERNAME,
```

```
  AIO_KEY
);

Adafruit_MQTT_Publish distanceFeed =
Adafruit_MQTT_Publish(
&mqtt,
AIO_USERNAME "/feeds/distance"
);

void MQTT_connect();

void setup()
{

  Serial.begin(115200);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  Serial.println("Connecting to WiFi");

  WiFi.begin(WLAN_SSID,WLAN_PASS);

  while(WiFi.status()!=WL_CONNECTED)
  {
    delay(1000);
    Serial.print(".");
  }

  Serial.println();

  Serial.println("WiFi Connected");

}

void loop()
{

  MQTT_connect();

  digitalWrite(TRIG_PIN,LOW);
```

```
  delayMicroseconds(200);

  digitalWrite(TRIG_PIN,HIGH);
  delayMicroseconds(1000);

  digitalWrite(TRIG_PIN,LOW);

  long duration = pulseIn(ECHO_PIN,HIGH);

  float distance = duration *0.0343/2;

  Serial.print("Distance : ");
  Serial.print(distance);
  Serial.println(" cm");

  if (distance < 20) {
    Serial.println("Status   : Intruder Detected");
  }
  else {
    Serial.println("Status   : Area Safe");
  }

  if (!distanceFeed.publish(distance)) {
    Serial.println("Cloud    : Upload Failed");
  }
  else {
    Serial.println("Cloud    : Data Uploaded");
  }

  Serial.println("----------------------------");

}

void MQTT_connect()
{

  int8_t ret;

  if(mqtt.connected())
  {
    return;
```

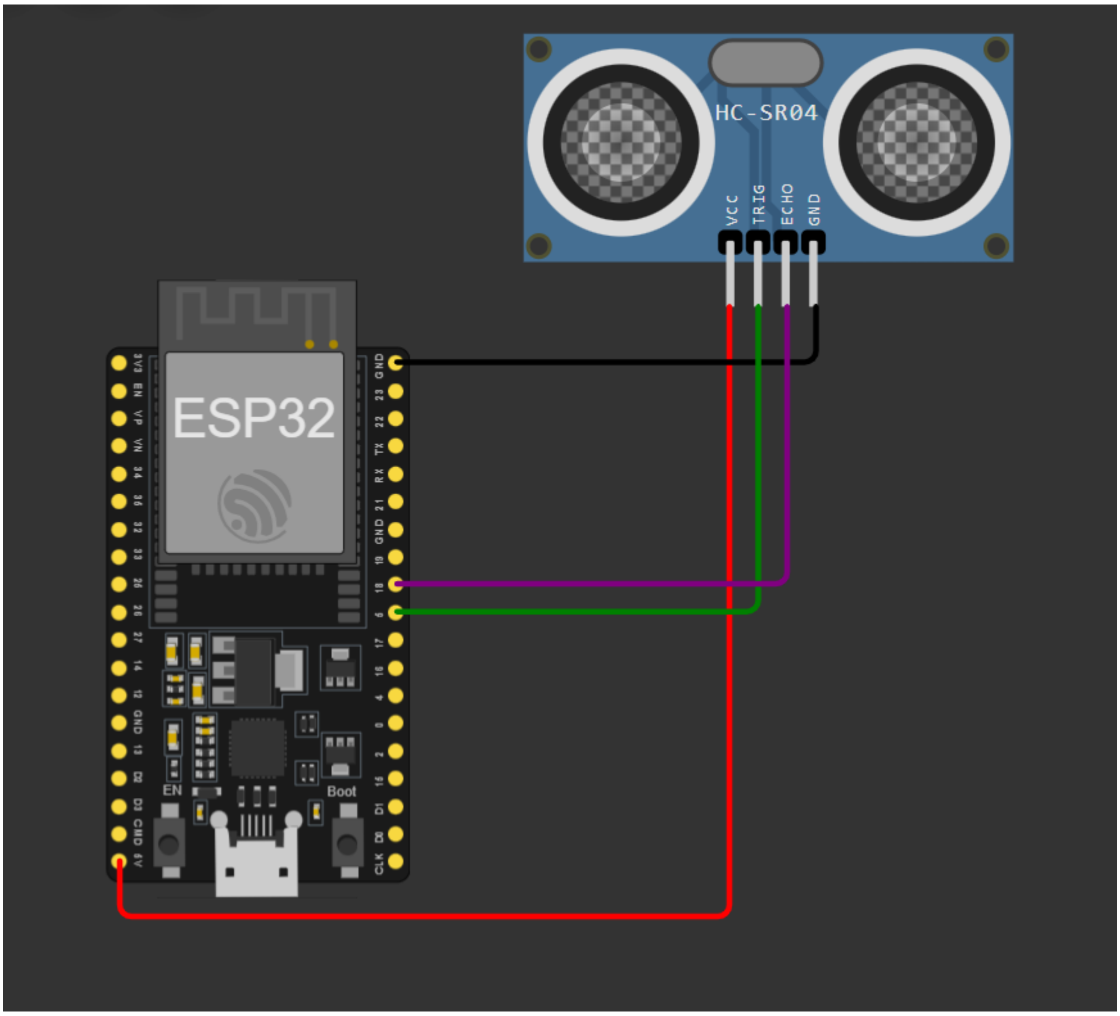
```
  }

  while((ret=mqtt.connect())!=0)
  {
    mqtt.disconnect();
    delay(5000);
  }

}
```

Results

Hardware Setup



Cloud Dashboard

Sagana127_ / Dashboards / Security Dashboard

Distance

```
Distance 403.52
Distance 403.49
Distance 403.45
Distance 403.47
Distance 403.42
Distance 403.52
Distance 403.44
Distance 403.47
Distance 403.42
Distance 403.47
Distance 403.45
Distance 403.49
Distance 403.44
Distance 403.45
Distance 403.44
Distance 43.34
Distance 43.36
Distance 43.36
Distance 43.17
Distance 403.44
Distance 403.47
Distance 403.49
Distance 403.42
Distance 403.42
```

Serial Monitor Output

```
WiFi Connected
Distance : 403.52 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 403.51 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 403.42 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 403.52 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
```

```
---------------------------
Distance : 403.44 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 43.17 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 43.36 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 43.34 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 43.36 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
```

```
---------------------------
Distance : 43.36 cm
Status   : Area Safe
Cloud    : Data Uploaded
---------------------------
Distance : 1.99 cm
Status   : Intruder Detected
Cloud    : Data Uploaded
---------------------------
Distance : 1.99 cm
Status   : Intruder Detected
Cloud    : Data Uploaded
---------------------------
Distance : 2.32 cm
Status   : Intruder Detected
Cloud    : Data Uploaded
---------------------------
Distance : 1.99 cm
Status   : Intruder Detected
Cloud    : Data Uploaded
---------------------------
```

Conclusion

The project successfully demonstrated cloud-connected IoT telemetry. The ESP32 measured distance, uploaded real-time data to the cloud, and reliably detected intrusions while providing live monitoring.